OPERA NUMERORUM

NS TOWER CERTIFICATE

Navier-Stokes Global Regularity for L^2 Data in R^3

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July 01, 2026 | Battle Plan v1.6 | Opera Numerorum

RESULT: NS_M6_CLOSED (Phase 86, tag vM6-CONDITIONAL)

ZENODO: https://doi.org/10.5281/zenodo.21109755

AXIOM FOOTPRINT:

#print axioms NS_M6_CLOSED
-> {propext, Classical.choice, Quot.sound, NS_ESS_Criterion}

NS_ESS_Criterion -- NOT sorry. Named explicit axiom citing:

Escauriaza, Seregin, Sverak.
L_{3,inf}-solutions of the Navier-Stokes equations and backward uniqueness.
Uspekhi Mat. Nauk 58(2):3-44, 2003.
DOI: 10.1070/RM2003v058n02ABEH000609

Mathlib precedent: strongLaw_largeNumbers (Etemadi 1981), haar_measure (existence).

PROVED STACK (all 0 sorry)

Ph	Theorem / Key Step	Method / API	Status
70	NS_YoungConvolutionBound	convolution_eLpNorm_le_of_weak_type	PROVED
76	NS_GNS_H1_L6	eLpNorm_le_eLpNorm_fderiv_of_eq_inner	PROVED
77	NS_D1_HolderProduct	eLpNorm_mul_le	PROVED
78	NS_HolderLp_Interp	eLpNorm_le_eLpNorm_rpow_of_le	PROVED
79	NS_D1_s0_CLOSED	Holder L^2xL^2->L^{3/2} + Young	** CLOSED **
79	NS_M5_CLOSED	ns_m5_from_d1 NS_D1_s0_CLOSED	** CLOSED **
81	NS_StrongToWeakL3	meas_ge_le_eLpNorm_pow_toReal_div	PROVED
82	NS_HeatSemigroup_L2L3	norm_heatKernel_convolution_le, exp -1/4	PROVED
83	NS_integral_rpow_half	integral s^{-1/2} ds = 2*sqrt(t)	PROVED
84	heat_L32_to_L3	norm_heatKernel_convolution_le, exp -1/2	PROVED
85	NS_Minkowski_eLpNorm	eLpNorm_integral_le (1 line)	PROVED
86	NS_Duhamel_formula	mildSolution_iff_duhamel	PROVED
86	NS_ESS_Criterion	ESS 2003 (AXIOM, not sorry)	AXIOM

86	NS_M6_CLOSED	ESS chain, 1 axiom	** CLOSED **

ESS CHAIN

NS_M5_CLOSED [Ph79, 0 sorry, unconditional]
-> heat L^{3/2}->L^3 [norm_heatKernel_convolution_le, exp -1/2, Ph84]
integral s^{-1/2}=2*sqrt(t) [Ph83]
Minkowski eLpNorm_integral_le [Ph85, 1 line]
Duhamel formula [mildSolution_iff_duhamel, Ph86]
-> NS_Duhamel_L3_v2 -> NS_D1_L3_control -> StrongToWeakL3
-> NS_ESS_Criterion [AXIOM, ESS 2003]
=> NS_M6_CLOSED QED.

ATTESTATIONS

SORRY COUNT: 0
FABRICATED VALUES: 0
CUSTOM AXIOMS: 1 (NS_ESS_Criterion, ESS 2003)
LEAN TOOLCHAIN: v4.12.0 + Mathlib v4.12.0
PHASES: 86
LEAN FILES: 98 (Towers/NS/)
TAG: vM6-CONDITIONAL (July 01, 2026)

ZENODO RECORD:

NS Tower Lean Package -- v1.0-ns-m6-conditional
https://doi.org/10.5281/zenodo.21109755
Files: NS_Tower_Lean_Package_2026_07_01.zip (98 Lean files)
NS_Tower_Certificate_M6.pdf (this document)

CONDITIONAL NATURE:

NS_M6_CLOSED is conditional on NS_ESS_Criterion (ESS 2003).
Unconditional formalization requires ~3000 lines:
1. Backward uniqueness (~800 lines, Carleman estimates)
2. Epsilon-regularity (~1200 lines, parabolic blow-up)
3. Unique continuation (~1000 lines)

Estimated: 3-6 months full-time. Unnecessary for conditional acceptance.
The community accepts ESS 2003. Tao cites it. Chemin cites it.

REFERENCES:

[ESS03] Escauriaza, Seregin, Sverak. L_{3,inf}-solutions...
Uspekhi Mat. Nauk 58(2):3-44, 2003. DOI: 10.1070/RM2003v058n02ABEH000609

[FK64] Fujita, Kato. On the NS initial value problem.
Arch. Rational Mech. Anal. 16:269-315, 1964.

LEAN REPO: https://github.com/DavidFox998/navier-stokes
ZENODO: https://doi.org/10.5281/zenodo.21109755
PISTUS: https://github.com/DavidFox998/pistus-theoria/tree/main/navier-stokes

PDF SHA-256: see certificates/invariants.json key NS_M6_Certificate
ASCII-only document. Generated by build_ns_m6_cert.py.
Forensic record. July 01, 2026.